

MEDICAL APPOINTMENT APPLICATION

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INTRODUCTION

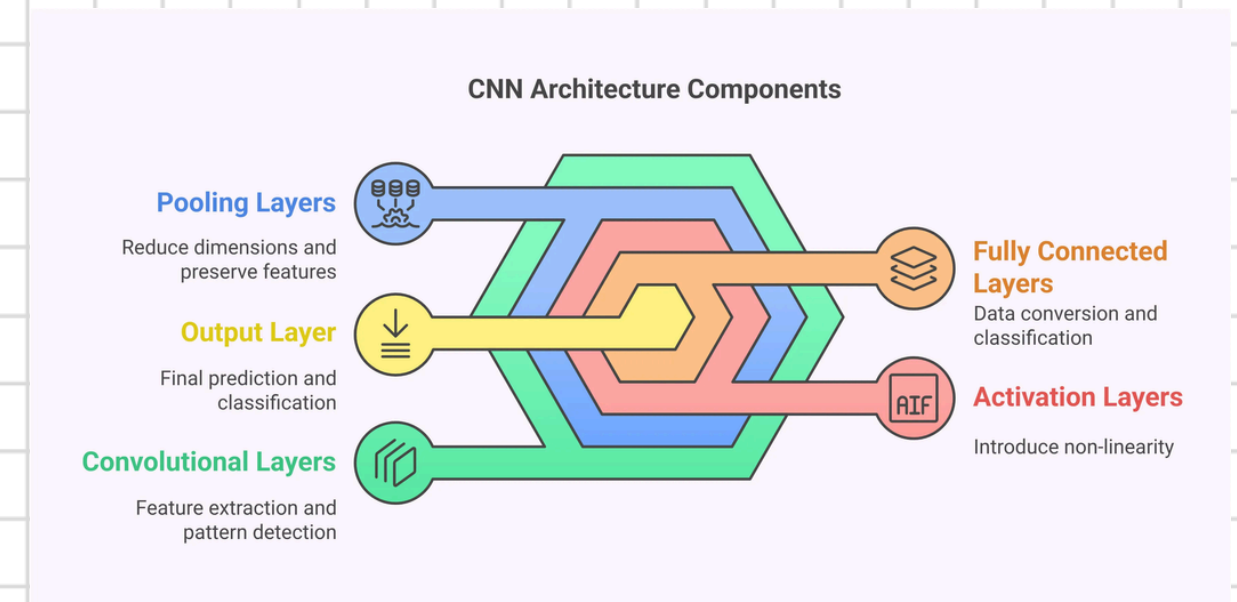
Motivation:

- In-memory databases (like Redis) are widely used for performance optimization but have become attractive targets for hackers.
- With weak configurations, Redis is highly vulnerable to attacks, specifically Redis Injection.
- This research builds a simulated lab environment to understand exploitation methods and practice Incident Response procedures safely

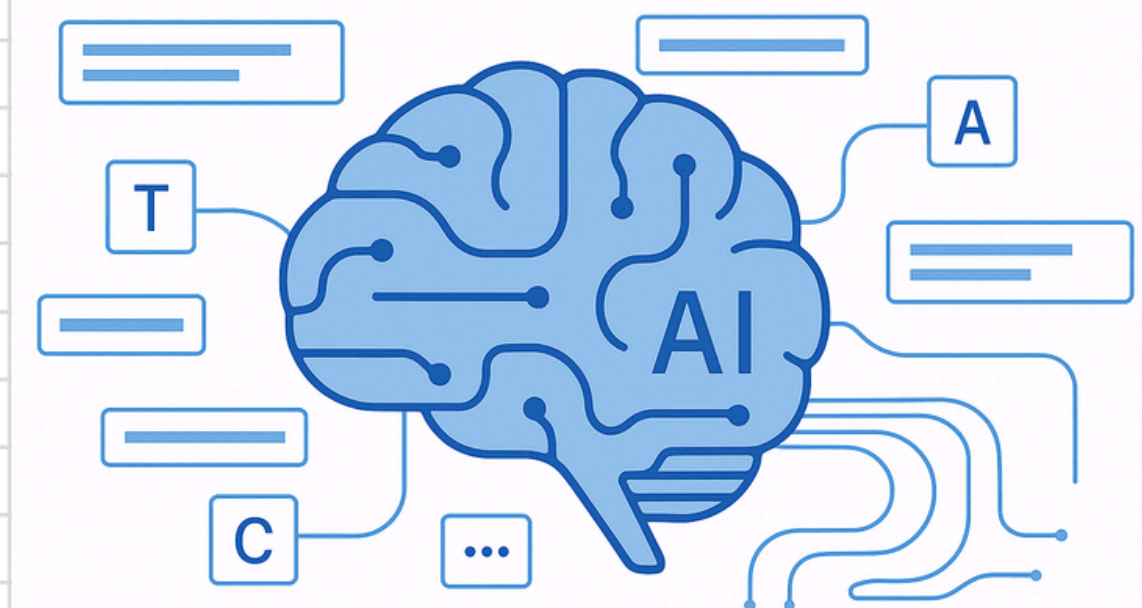


AI Theory

- **AI overview:** Artificial Intelligence (AI) is a field of computer science focused on enabling machines to simulate human intelligence such as learning, reasoning, and decision-making.
- **CNN model:** A Convolutional Neural Network (CNN) is a deep learning model widely used for image processing and computer vision tasks. It operates by extracting spatial features through convolution layers.
- **LLM model:** A Large Language Model (LLM) is an AI model trained on massive text datasets to understand and generate human language.



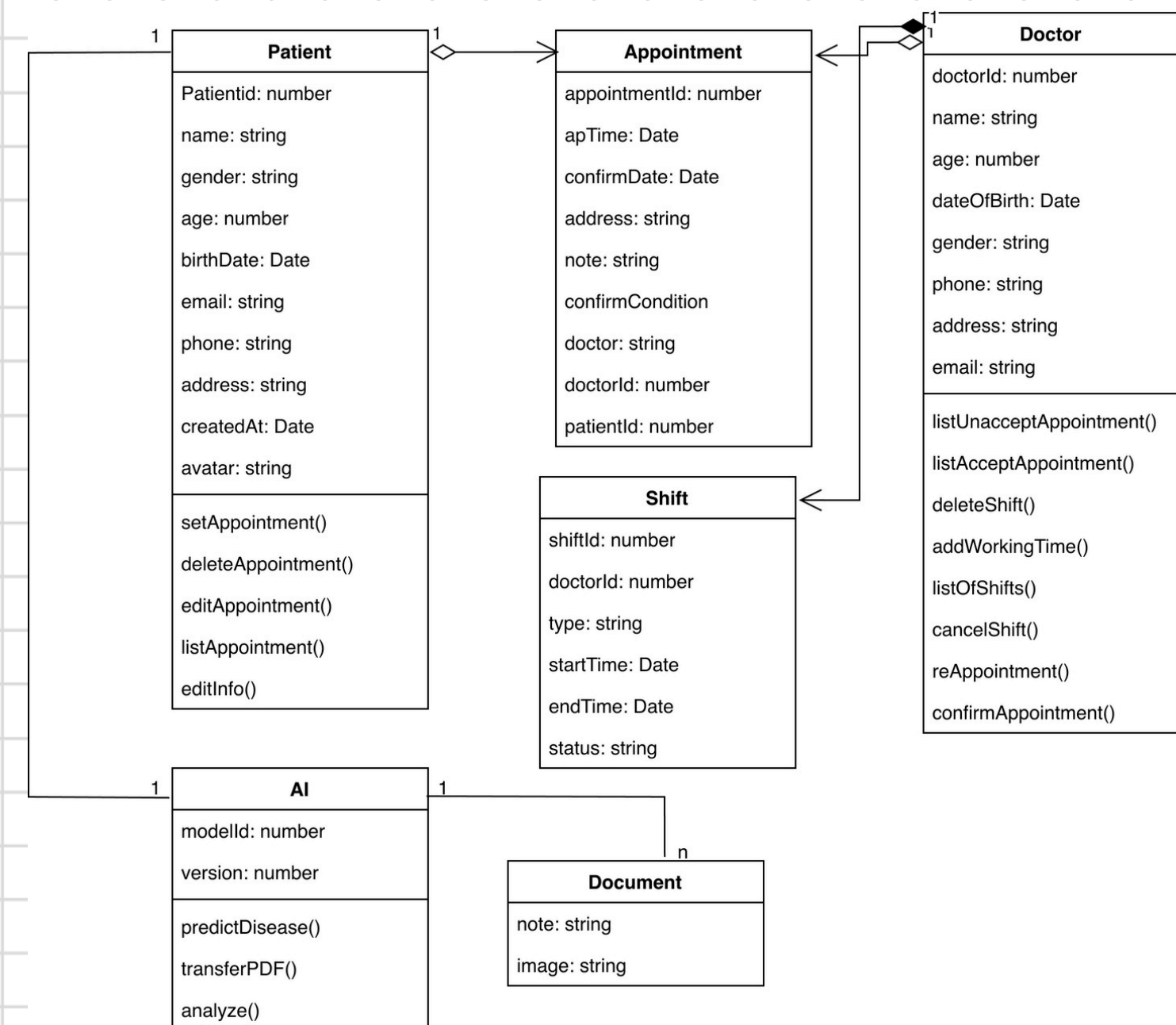
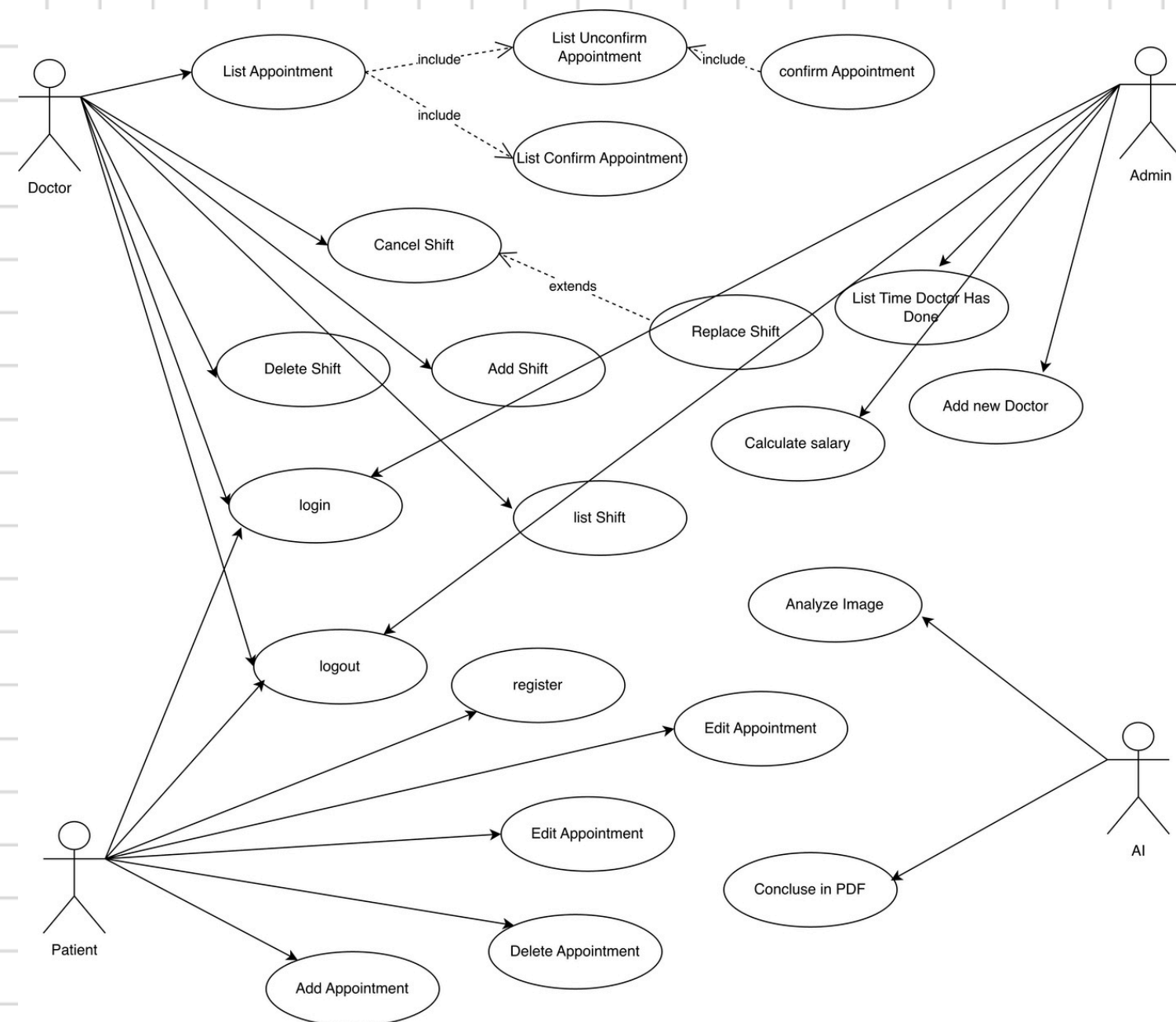
LARGE LANGUAGE MODEL



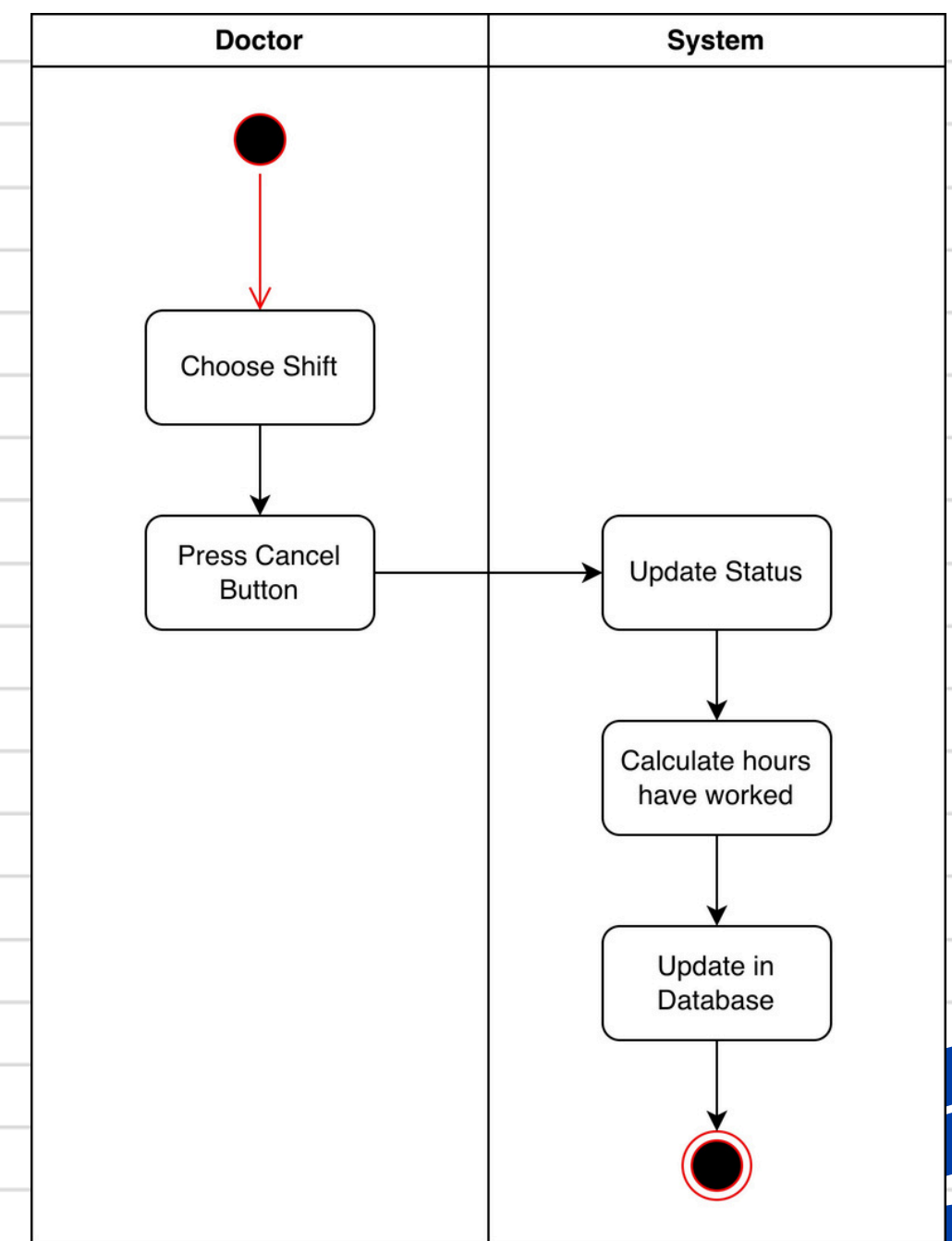
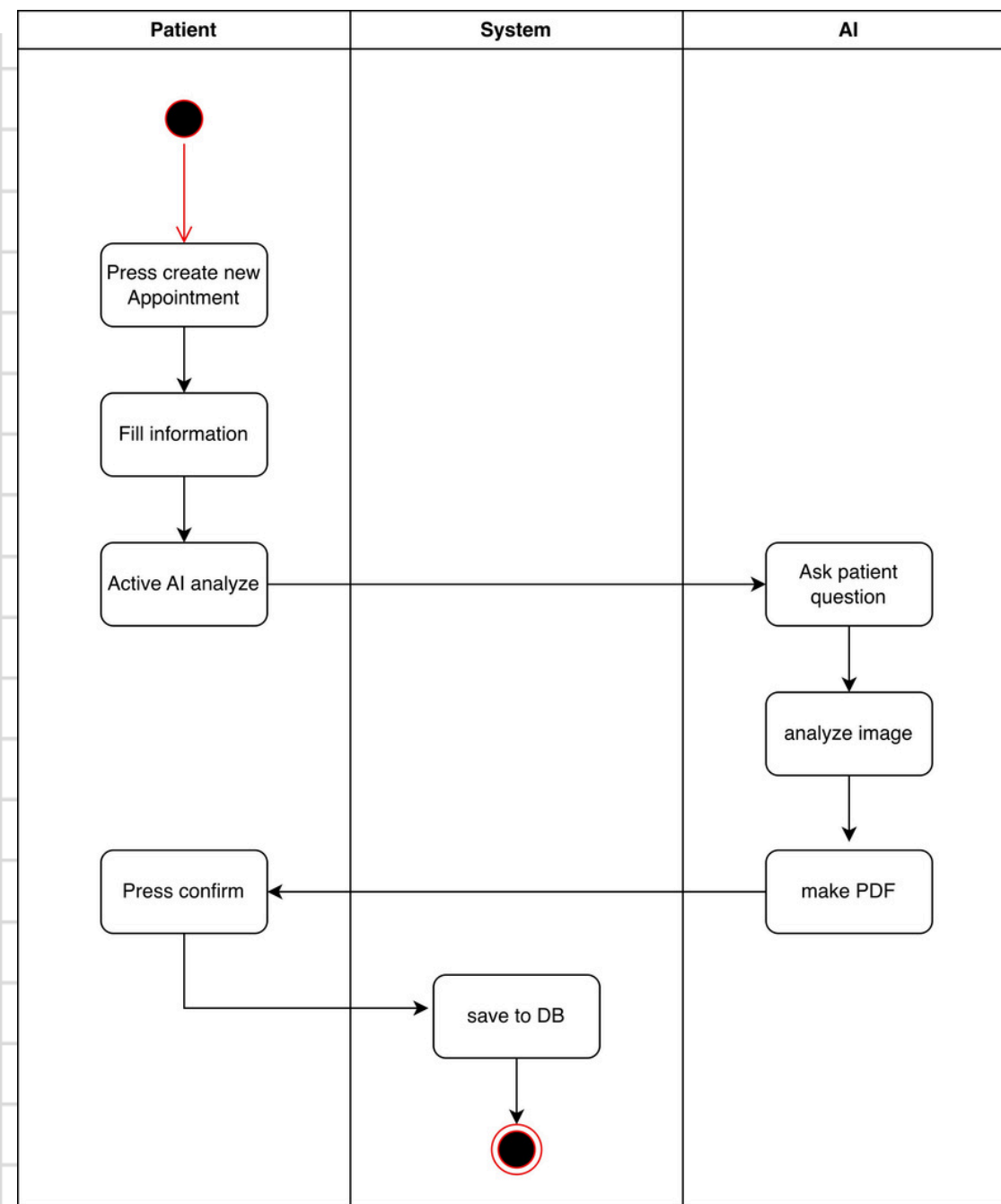
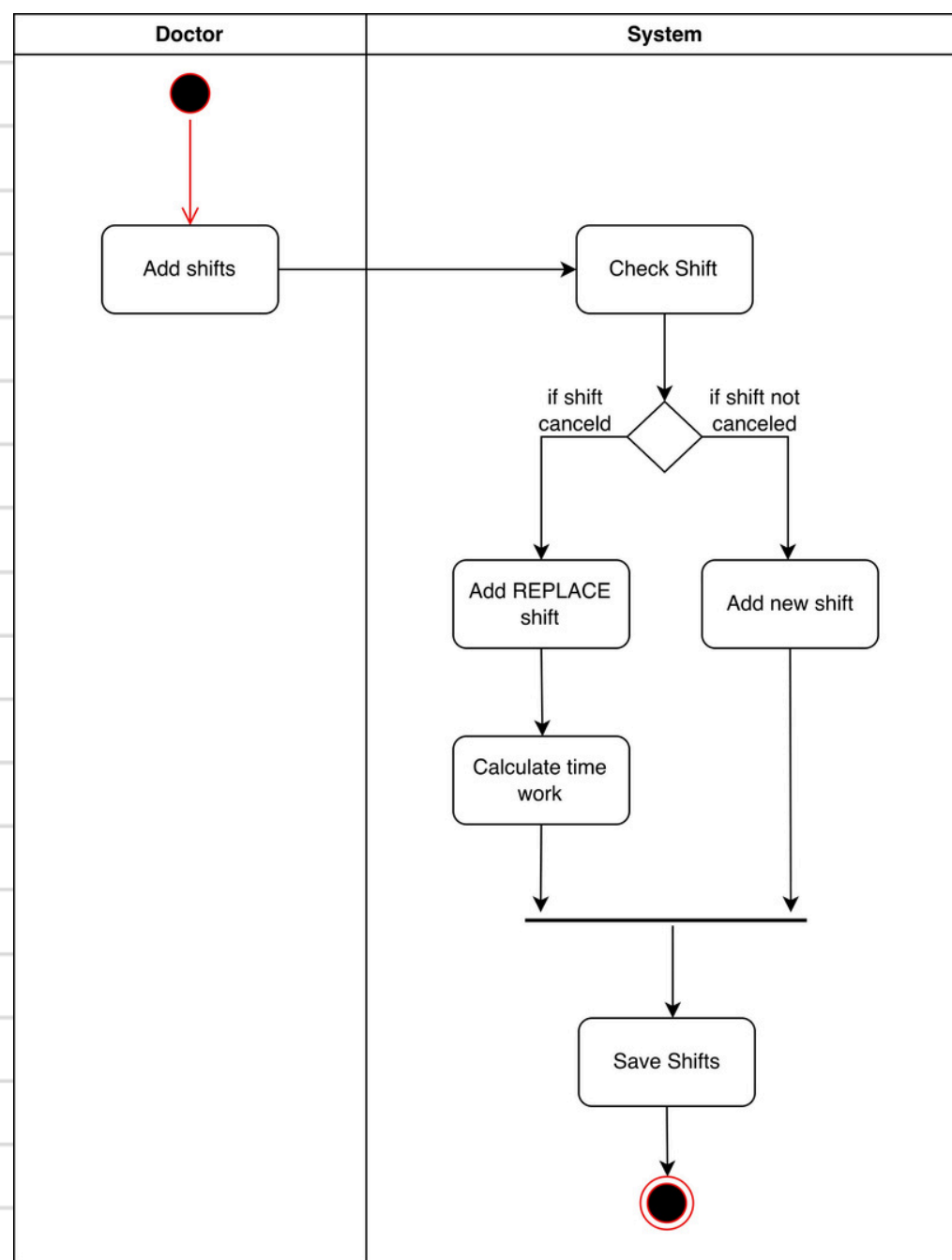
System Analysis and Design

- Use-case diagram: The use case diagram illustrates the interactions between users and the system
- Class diagram: The class diagram represents the main system structure, including entities such as User, Patient, Doctor, Appointment, Schedule, and Notification
- Activity diagram: The activity diagram shows the workflow of booking an appointment. A patient logs into the Flutter app, searches for a doctor, selects an available time slot, and submits a booking request
- Sequence diagram: The sequence diagram illustrates the interaction between Patient (Flutter app), API Gateway (NestJS backend), Database, and Notification service.

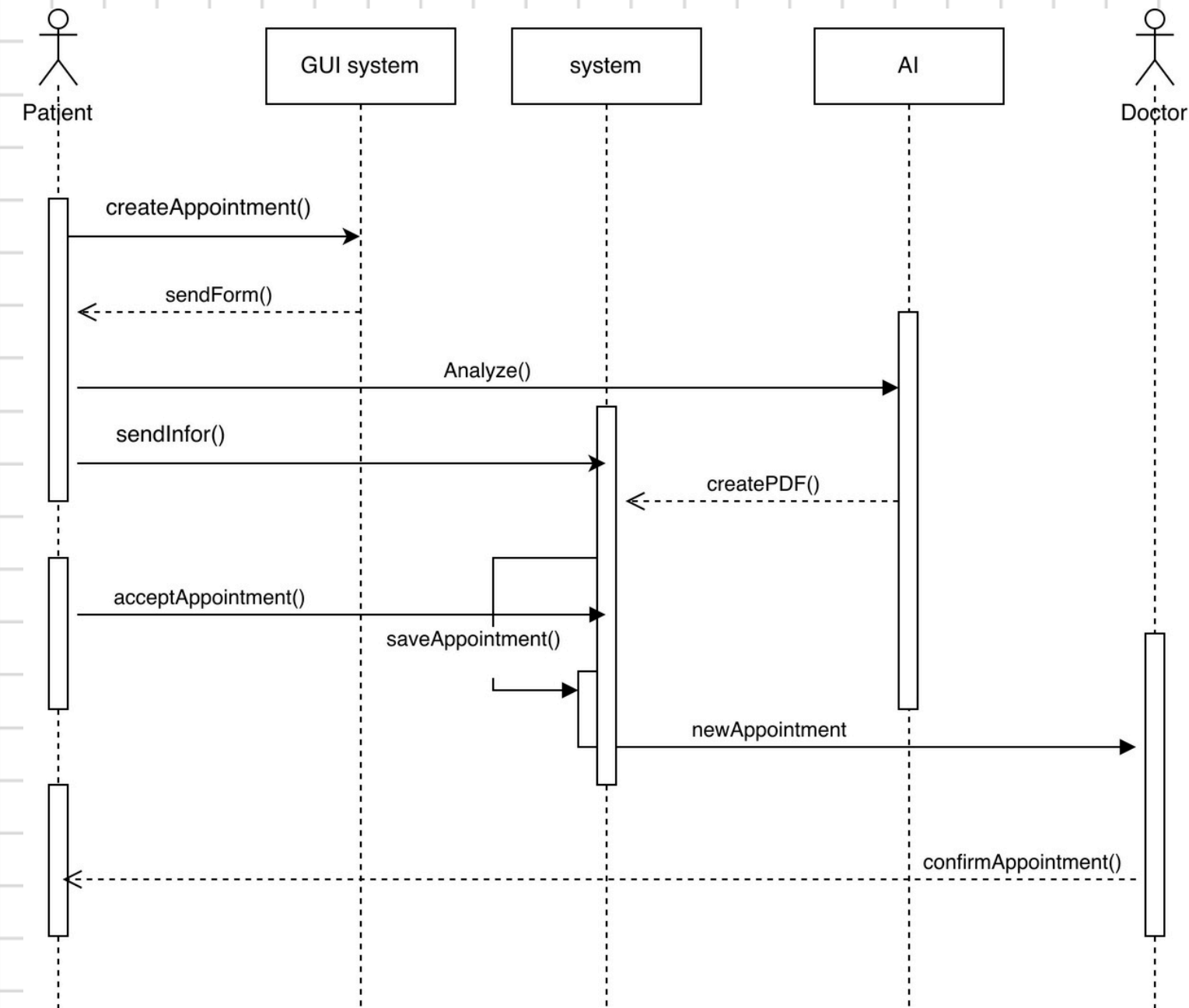
System Analysis and Design



System Analysis and Design

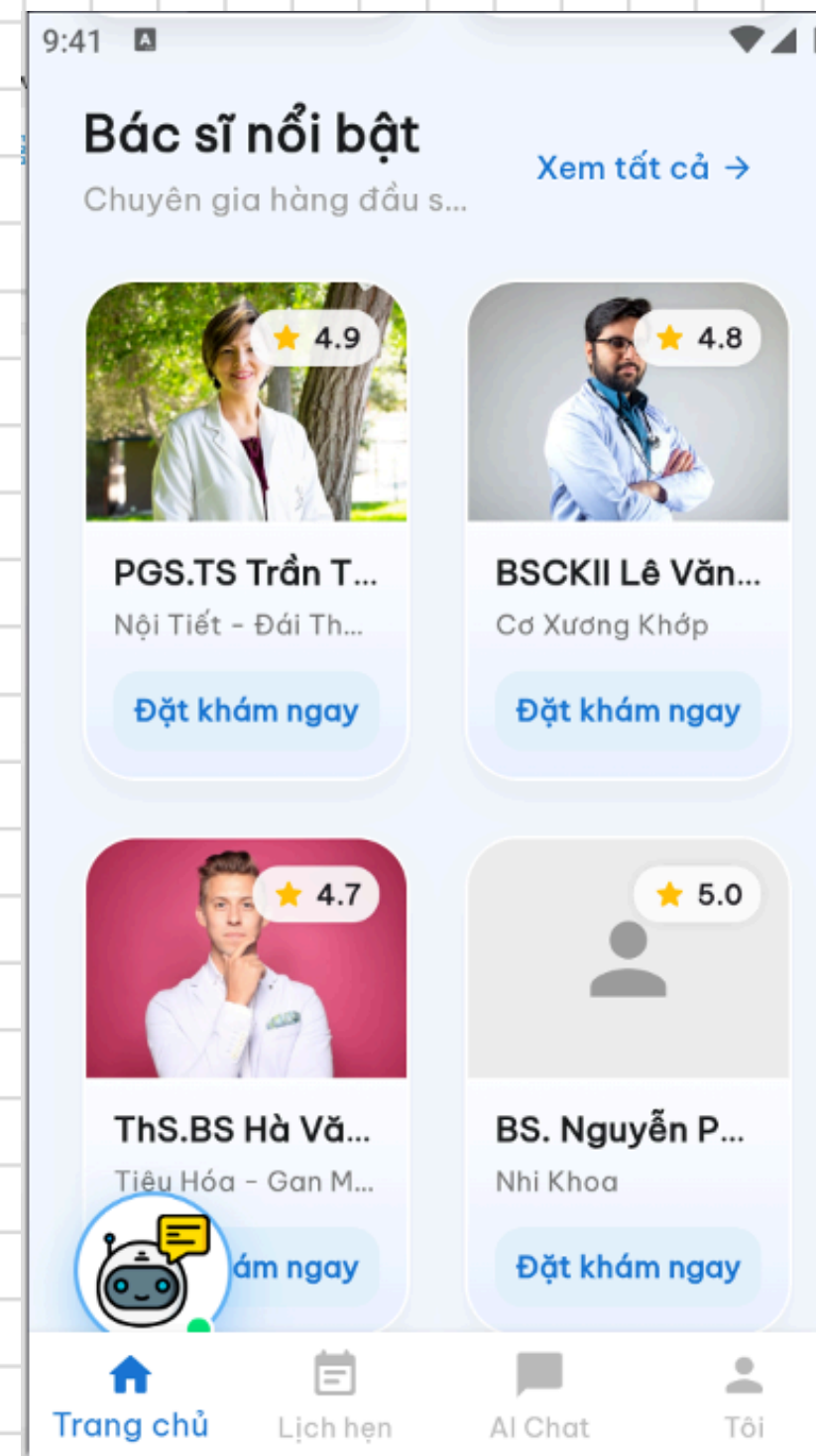
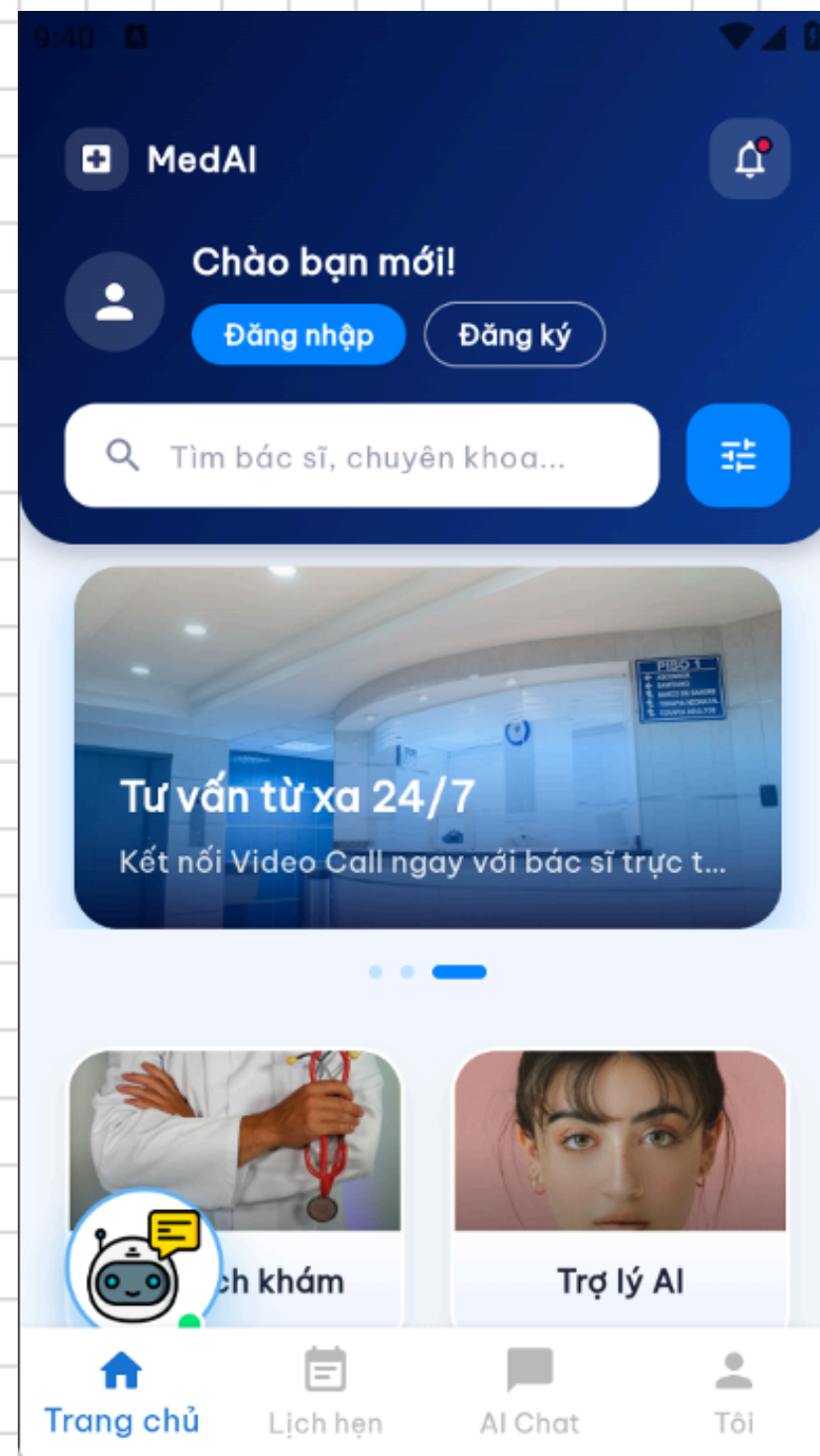


System Analysis and Design



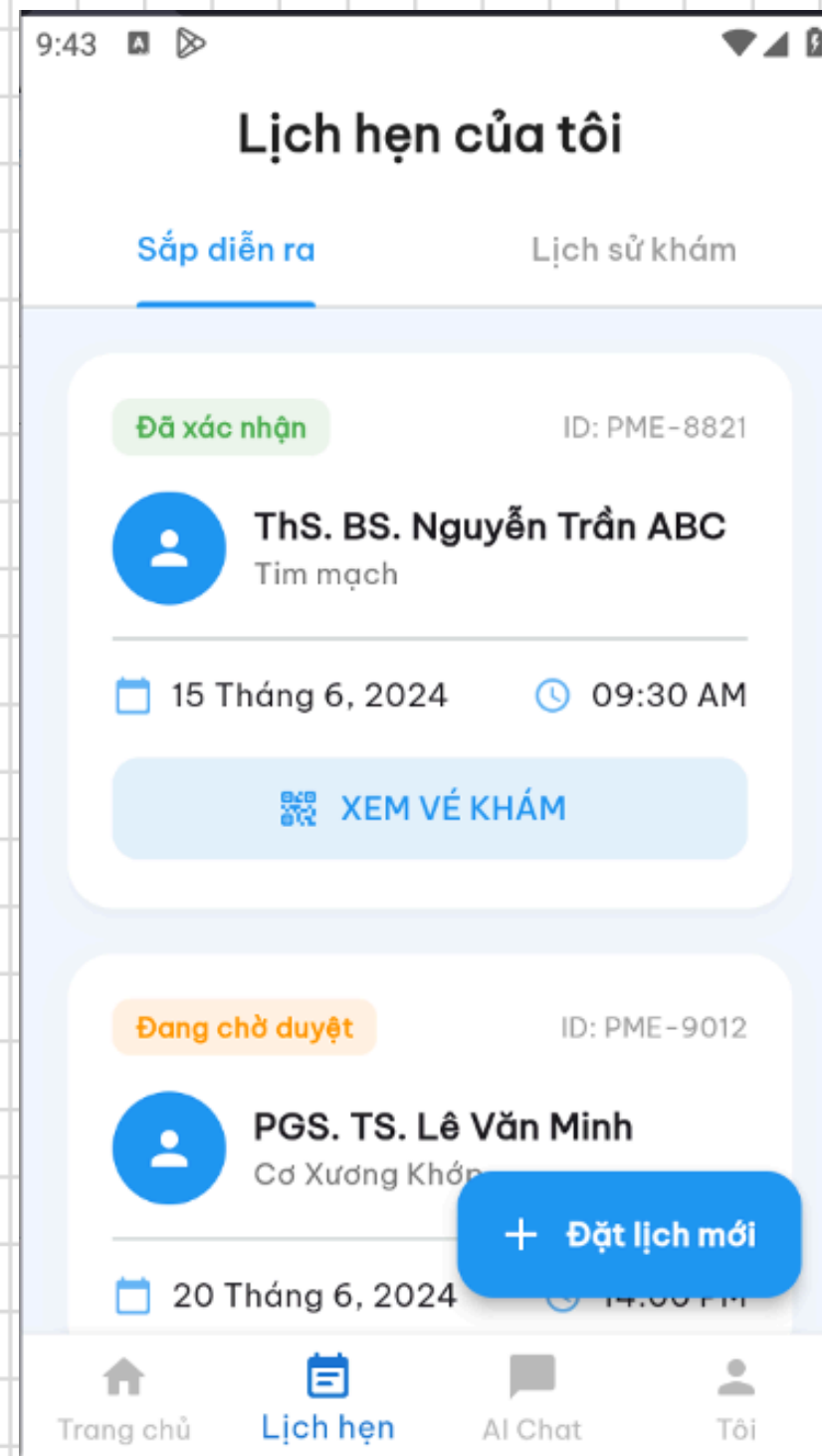
Visualize of the project

Patient page



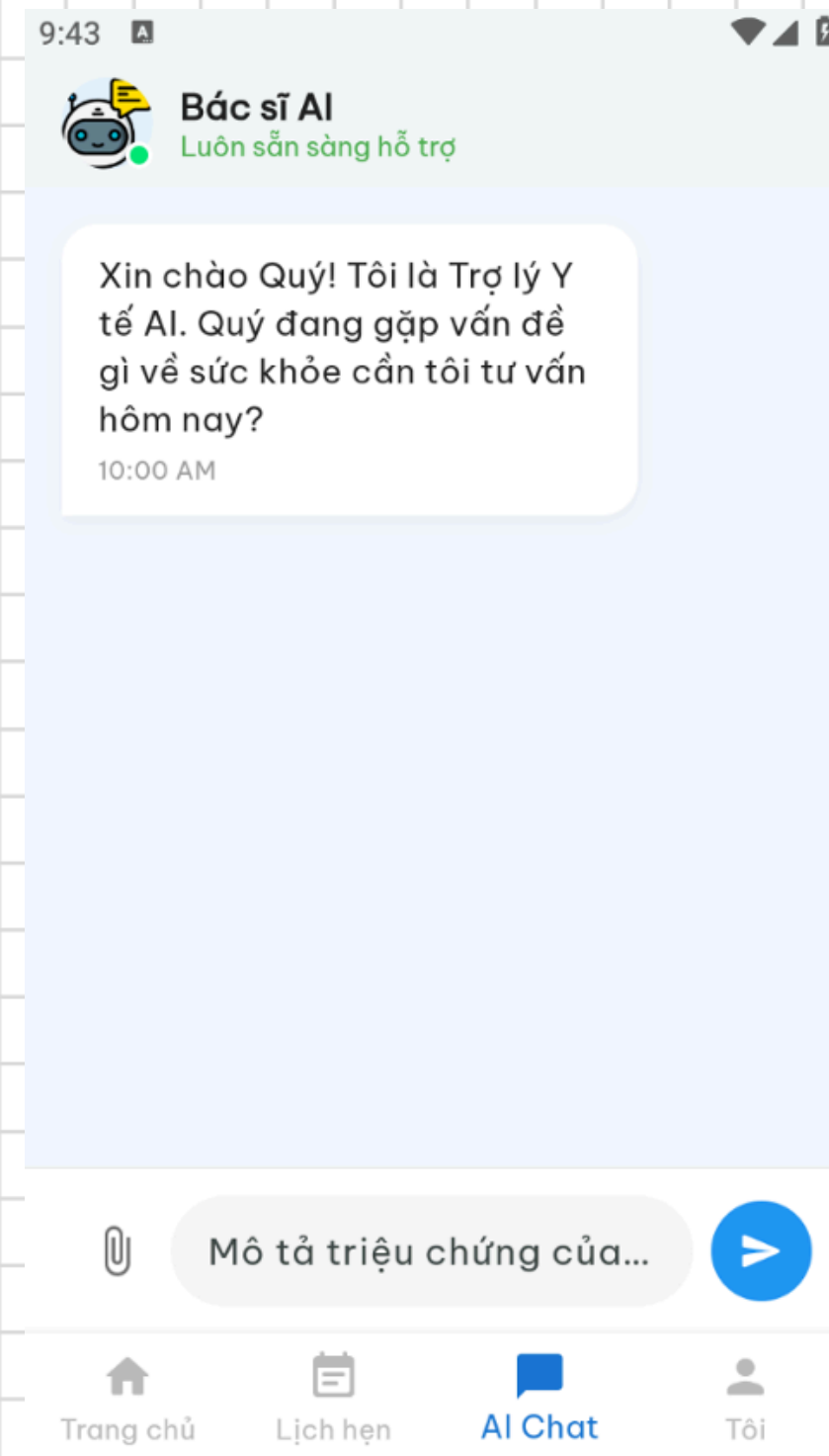
Visualize of the project

Patient page



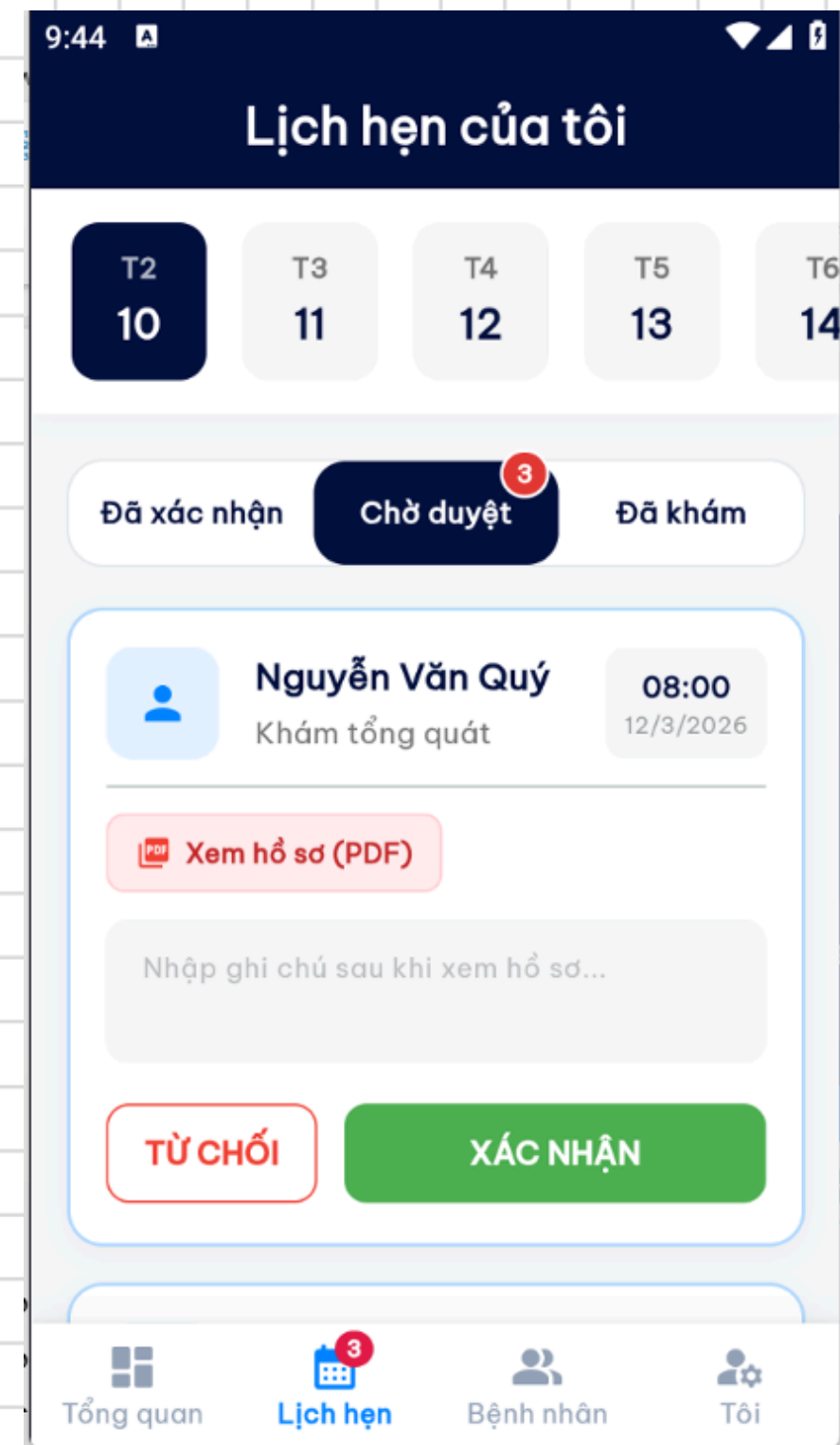
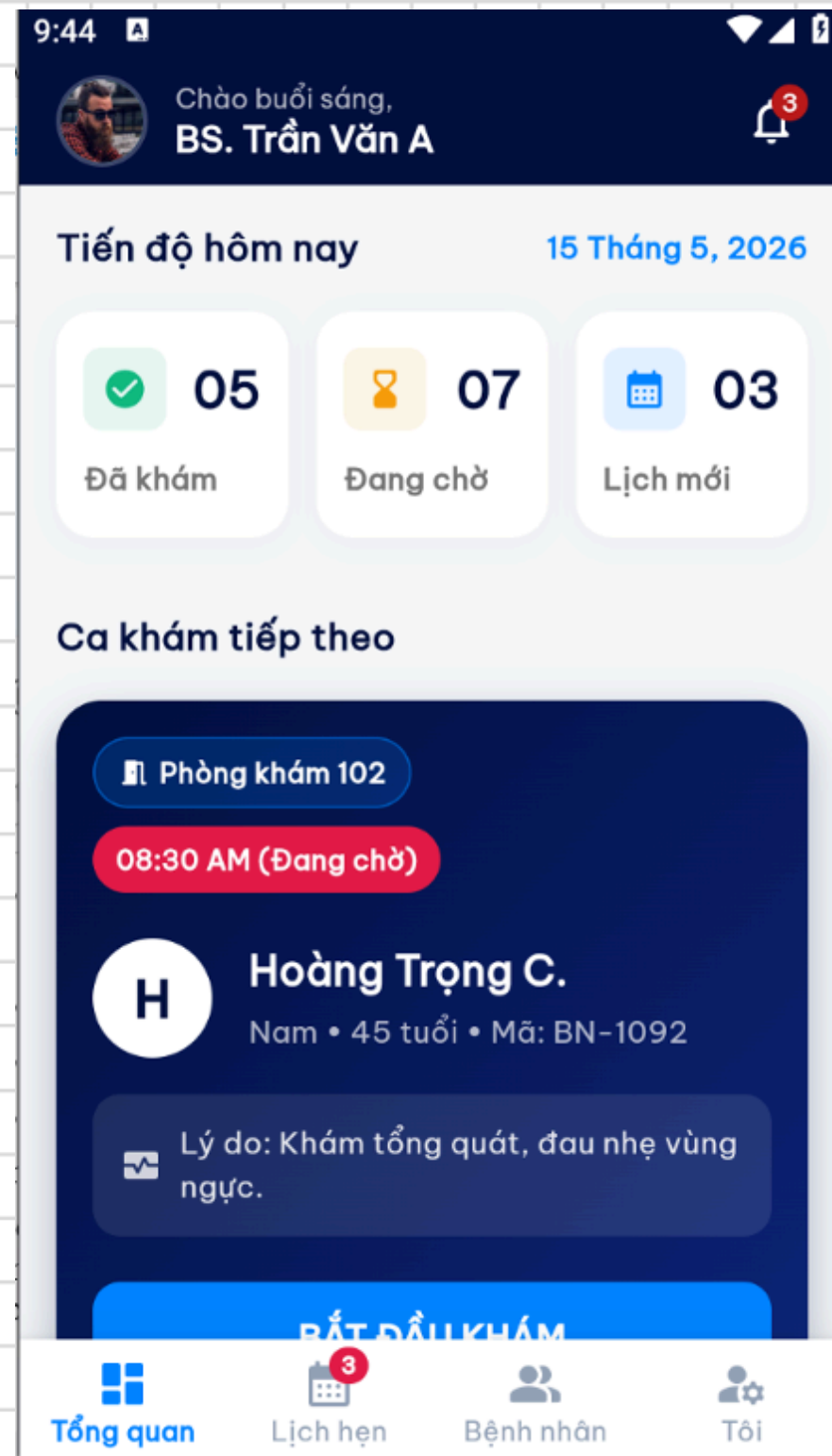
Visualize of the project

AI assistance



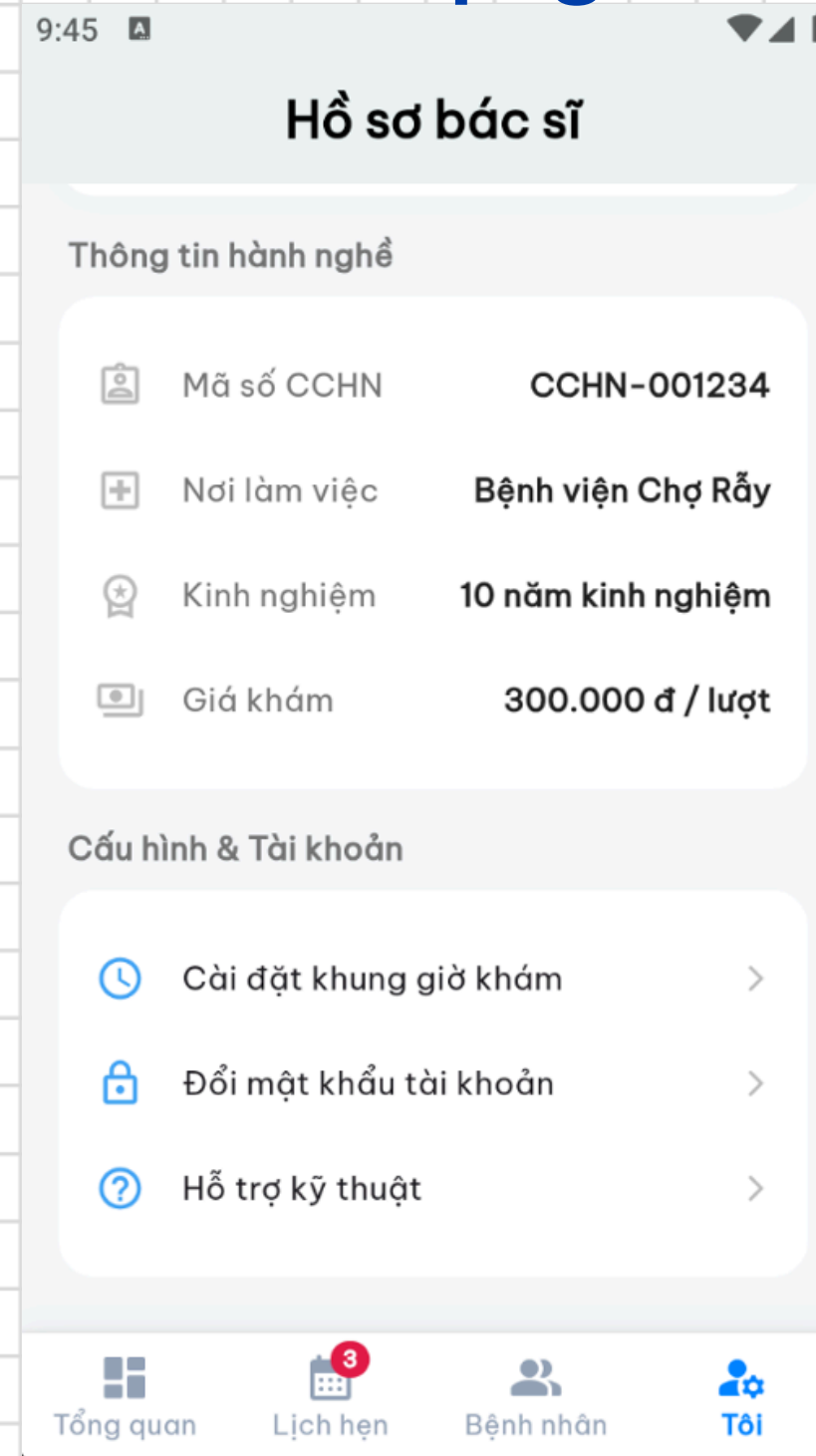
Visualize of the project

Doctor page



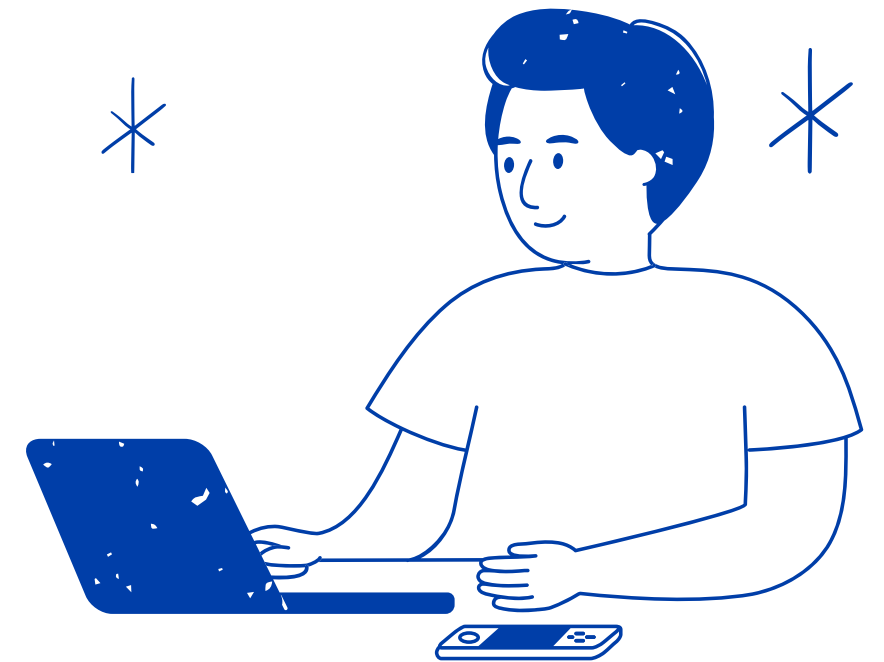
Visualize of the project

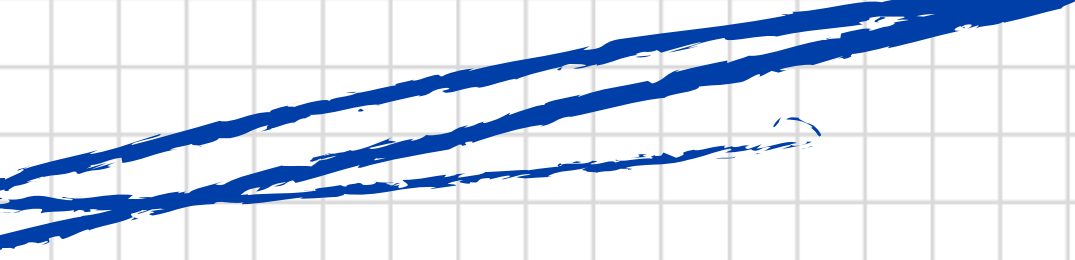
Doctor page



Conclusion & Future Directions

- **Key Achievements**
 - **AI Booking System:** Developed a medical appointment app using NestJS and Flutter with basic AI support for doctor recommendations.
 - **Cross-Platform App:** Built a working mobile app for both patients and doctors with smooth booking features.
- **Limitations**
 - **Scope:** Limited to core booking features and simple AI due to time constraints.
 - **Data:** AI performance depends on limited training and usage data.
- **Future Directions:**
 - **AI Improvement:** Enhance recommendation accuracy and personalization.
 - **UI/UX:** Improve user experience for both patients and doctors.
 - **Scalability:** Expand to multi-clinic and hospital-level systems.





DEMO

